



ATENEO DE DAVAO UNIVERSITY
SCHOOL OF ENGINEERING AND ARCHITECTURE



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DOCUMENTATION REPORT
THE WISH-CYCLING CRISIS

SUBMITTED TO:
GABRIEL QUIBOD
PROFESSOR

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SUBMITTED BY:
KYLLE CHRISTIAN L. BARREDA
BSCE 3C
STUDENT

The world is producing more trash than ever, but our recycling systems are failing. The real-world problem isn't just that we have trash; it's that we don't know how to sort it correctly. When "good" recycling is mixed with food or non-recyclables, the whole batch ends up in a landfill. This is relevant because landfills are reaching capacity and polluting our environment. This affects everyone, from local governments struggling with waste costs to everyday citizens living in increasingly polluted areas.

The core issue is "Wish-cycling"—when people throw items into recycling bins hoping they are recyclable, without actually knowing if they are. Sorting rules are confusing and change by location. Is this coffee cup paper or plastic? Does the lid stay on? Most people guess, leading to high contamination rates. Studies show that up to 25% of items placed in recycling bins are actually contaminants. Human Error: Sorting is tedious and confusing. There is no immediate "reward" for recycling correctly, so people lose interest over time.

We propose an AI-powered mobile application designed to remove the guesswork from waste disposal and turn a chore into a rewarding habit.

AI Vision Sorter: Users simply point their phone camera at an item. The AI instantly identifies the material (e.g., "HDPE Plastic" or "Compostable Paper") and tells them exactly which bin it belongs to.

Smart Disposal Verification: To prevent cheating or mistakes, the app uses the camera to verify the item was actually dropped into the correct bin before granting points.

